Kinds as Projectibility Profiles

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Abstract

Anti-essentialist theories of kinds explain how categories can be real without sharing an essence. Homeostatic property cluster theory gives the central Boydian answer, but HOMEOSTATIC now covers several different achievements: stable projectible profiles, network order, maintenance, and corrective control. This paper proposes a more portable vocabulary. Kinds are projectibility profiles: worldly property-and-relation patterns whose partial observation licenses specified inferences under specified conditions. The analysis of a profile records its diagnostics, the relations that stabilize it, its limits, and failure modes. Stabilizing relations and mechanisms matter because they explain why those projections work. Homeostatic control is a demanding subcase, not the default form of kindhood.

Keywords: natural kinds; projectibility; homeostatic property clusters; classification; philosophy of science

1 Introduction

Anti-essentialist kind theory begins from a familiar pressure. Many categories that matter to science, grammar, social inquiry, and ordinary reasoning don't share an essence. Their members needn't have one property in common, and their boundaries may be graded, historically contingent, or context-sensitive. Still, some of these categories support reliable inference. They let us predict, explain, intervene, measure, compare, or learn.

Boyd's homeostatic property cluster framework made that combination possible. A kind can be real when a cluster of properties supports induction and explanation because classificatory practice has been accommodated to worldly structure rather than because every member shares a defining essence (Boyd, 1991, 1999). That insight remains the starting point. But the word HOMEOSTATIC has been asked to do too much. In different contexts it can name a stable profile, a network of relations, a maintaining process, or a corrective-control system.

Those are not the same achievement. A profile can be stable without a known maintainer. It can be ordered without being maintained. It can be maintained without being controlled. And it can support useful projection without earning the stricter label HOMEOSTATIC. Slater's stable property

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cluster account presses the stability point (Slater, 2015); Khalidi's causal-network account presses the order point (Khalidi, 2013, 2018). The lesson is not that Boyd was wrong, but that the evidential work needs sharper labels.

This paper proposes one such label: PROJECTIBILITY PROFILE. A profile, in this use, is the worldly property-and-relation pattern whose partial observation licenses specified inferences under specified conditions. The analysis of a profile specifies that pattern: which projection it supports, how we diagnose it, what makes the projected-from and projected-to properties co-available, what stabilizes that co-availability, and where the projection fails.

I use KIND inclusively for any projectibility profile. A NATURAL KIND claim and a HOMEOSTATIC claim are stronger, separable claims about the profile's support. This deliberately shifts the discriminating work from a single kind/non-kind boundary to tiers of support.

The priority of projectibility is methodological and explanatory. We start by asking what the category lets us infer, then ask what profile supports that inference, what relations make the profile non-accidental, what stabilizes it, and where it fails. The order of inquiry doesn't imply that projectibility is metaphysically prior to every stabilizing structure. It says that a mechanism matters to kind theory because it explains projection, not because mechanism talk by itself makes a category real.[1]

The contribution is regulative rather than metaphysical in the grand style. It doesn't introduce a new kind of entity beyond Boydian profiles, Slaterian stability, or Khalidian networks. It gives those resources an order of use: name the projection, specify the profile, identify the support, and state the conditions under which the kind claim should be demoted, localized, or rejected.

The paper is deliberately synthetic. Section 2 defines the profile package and distinguishes profiles, stabilizers, maintainers, and controllers. Section 3 explains why the projection question comes first. Section 4 shows how the vocabulary travels across explanatory levels without turning every case into a biological homeostasis case. Section 5 gives the vocabulary teeth by making demotion and failure modes explicit. Section 6 states the payoff.

2 The Profile Package

A profile specification answers six questions. First, what's the projection target? Second, what pattern of properties and relations supports it? Third, what evidence diagnoses that pattern? Fourth, what ordering relation makes source and target properties co-available rather than accidental co-occurrences? Fifth, what stabilizes that co-availability across relevant contexts, timescales, or perturbations? Sixth, what scope conditions and failure modes mark the projection's limits?

The first question matters because recurrence is cheap. A list of co-occurring properties becomes theoretically interesting only when it licenses some further expectation. The expectation may be predictive, explanatory, classificatory, or intervention-guiding. It may concern a biological process, a grammatical pattern, a measurement practice, or a social role. But without a declared projection target, the category remains a description of observed association.

The second question asks what the profile is. A profile is the property-and-relation pattern whose partial observation licenses specified inferences. In a biological case, it may include morphology, development, ecology, and reproductive relations. In a grammatical case, it may include distributional behaviour, semantic tendencies, diachronic sources, acquisition patterns, and mismatch residue. In a

[1] This paper doesn't take a stand on the stronger constitutive thesis sometimes associated with PROJECTIBILITY-FIRST formulations. Nothing in the argument depends on that stronger claim.

social case, it may include institutional marks, practical responses, uptake conditions, sanctions, and repair paths. The pattern needn't be one-level or one-mechanism.

The third question concerns evidence. Diagnostics belong to the specification, not to the profile itself. They're the tests, traces, contrasts, measurements, or judgments by which inquiry locates the profile and checks whether a proposed projection travels beyond the cases that introduced it.

The fourth question asks why the profile is non-accidental. Khalidi's causal network account supplies one important model: kinds are not mere concatenations of properties, but nodes in ordered networks where some properties help explain others (Khalidi, 2018). In other domains, the relevant order may be functional, conventional, normative, or algorithmic rather than narrowly causal. The question is synchronic or structural: why are the projected-from and projected-to properties co-available in the cases at issue? In the broader use, the ordering relation still has to explain the declared projection; mere cue status isn't enough. Where the stronger Khalidian natural-kind claim is at stake, conventional or normative order needs causal, causal-historical, or causal-normative support. The level has to be marked. Otherwise a social coordination pattern, a cognitive mechanism, and a corpus distribution begin to explain one another by equivocation.

The fifth question is modal or diachronic. It asks what keeps, regenerates, or preserves the profile across the relevant range of contexts, timescales, and perturbations. Some answers to the fourth question also answer the fifth, but the claims differ. Network order says why co-availability isn't coincidence; stabilization says why it persists, recurs, or returns within the scope of the projection. Sometimes no separable stabilizer has been identified. Robust stability can still support limited projection, but stronger claims about maintenance or control require further support.

A STABILIZER explains why co-availability persists, recurs, or returns across the declared range. It may be a relation, constraint, mechanism, practice, or shared standard. Stabilizers needn't be active processes: selectional constraints, institutional practices, and ecological regularities can all stabilize a profile. What matters is that they explain why observing part of the profile gives reason to expect the rest.

A MAINTAINER is more specific. It's an active stabilizer that preserves or regenerates the profile over time or under perturbation. A maintainer may be developmental, social, institutional, biological, or interactional. It can keep a profile available without registering each departure as an error or returning a system to a fixed value.

A CONTROLLER is more specific again. It's a maintainer with perturbation-sensitive organization that preserves a higher-scale relation through lower-scale variation. On the stricter use defended in Reynolds (2026a), the term HOMEOSTATIC belongs here. Homeostasis is not a synonym for recurrence, order, or maintenance. It marks the demanding case where corrective organization preserves a relation despite perturbation.

One contrast keeps the levels apart. A valley's shape may be stabilized by erosion and gravity; wound closure is maintained by active healing; body temperature is controlled by corrective feedback. Only the last case is homeostatic in the strict sense.

The sixth question concerns scope and failure. A profile is projectible under some range of conditions, contrasts, and purposes. This doesn't make kindhood arbitrary. Purposes affect which available projections matter for an inquiry; they don't create the availability of those projections. A profile may support one family of inferences and fail for another. That gives us a reason to localize the claim, not a reason to deny realism wherever interests enter.

3 Why Projectibility Comes First

Goodman's new riddle made PROJECTIBILITY the name for a hard problem: some predicates that fit observed cases still don't license projection to unobserved ones (Goodman, 1955). The present use of PROJECTIBLE is Boydian rather than Goodmanian in one central respect. It doesn't treat projectibility primarily as entrenchment in our predicate-using history. It asks what worldly order makes a category a reliable guide to further expectation.

That realist turn is the point of Boyd's anti-essentialism. We don't need a shared essence to explain why a category supports induction. We need a non-accidental profile that our classificatory practice has partly learned to track (Boyd, 1991). Projectibility is therefore the target of the kind claim. Mechanisms, networks, histories, standards, and practices matter because they explain why projection works. In broadly Boydian terms, the projection target states the programmatic role; the profile specification gives the candidate explanatory support.

This order blocks a common slide. Recurrence gets promoted to stability, stability to stabilization, stabilization to homeostasis, and homeostasis to kindhood. The profile vocabulary reverses that order. Start with the inference. Ask what profile licenses it. Then ask which stabilizing claims have actually been earned.

Starting with a mechanism makes the slide easier. Once a process has been named, the temptation is to let it certify the kind before asking what projection it improves. The profile-first order forces the mechanism to answer a narrower question: what inference would become weaker if that mechanism were removed?

The point is not that mechanisms are secondary in the world. During inquiry, we may have good evidence that a profile is projectible before we know what maintains it. The reliable projection still depends on whatever worldly relations make the profile non-accidental. The epistemic order and the grounding order can come apart. The proposed discipline is to avoid treating the unknown ground as if it had already been identified, named, and promoted to homeostasis.

This also explains how stable property clusters fit the account. Slater is right that natural kindness needn't wait for a named homeostatic mechanism, and his stability demand is already projectibility-shaped: observing enough of a subcluster should license expectations about the larger cluster (Slater, 2015). Stability can be enough for a scoped kind claim when it licenses the declared projection. The profile question then asks which stronger claims about stabilizers, maintainers, or controllers have also been earned. A category may support modest local projections while failing to support broader ones. Both results are informative.

Purpose-relativity enters here with care. A profile is projectible for a family of questions, contrasts, and evidential practices. In physiology, a useful profile may license intervention and measurement. In grammar, it may license expectations about distributions, manipulations, and borderline cases. In social ontology, it may license expectations about uptake, status, sanctions, or institutional repair. The practice selects which projection matters. It doesn't make the projection true.

A projectibility profile therefore differs from a checklist. Checklists record features. Profiles connect features to expectations under scope conditions. A checklist asks whether an item has properties P_1, P_2, and P_3. A profile asks whether observing P_1 and P_2, in the relevant setting, licenses expectation of P_3, and what would defeat that expectation.

The difference matters for explanatory economy. A mechanism-first paper can finish by naming structures that hold a cluster together. A profile-first paper has to say what those structures let us project, what failures would demote the claim, and whether the advertised stabilizer operates at the right explanatory level. The contribution is a discipline for using categories as guides to further expectation.

This gives a simple test for mechanism talk. If removing the mechanism leaves the same predictions in place, the mechanism has not earned its role in the kind claim. It may still be a true description of the system, but it's not yet doing the explanatory job a projectibility-profile analysis needs.

4 Profiles Across Levels

The profile vocabulary is meant to travel, but not by making every domain look biological. Profiles can be behavioural, algorithmic, corpus-distributional, model-internal, institutional, normative, social-practical, or biological. The level matters because evidence at one level rarely licenses projection at another without a bridge claim.

Linguistic categories make the point concrete. In the deitality analysis of English, the point of introducing the category isn't to rename every definite form. The category captures a structural profile that supports further expectations: which diagnostics should cluster, which semantic mismatches should leave distributional residue, which contrasts should shift under prosody or specificity manipulation, and which variation should appear across dialects and acquisition (Reynolds, 2025a). The category earns its place by those projections.

The same discipline applies to typological comparison. A comparative concept can remain an instrument for measurement. It becomes a naturalized kind claim only if the comparandum shows a profile that travels across languages, supports out-of-sample projection, and comes with demotion criteria when the profile proves too thin (Reynolds, 2025b). Here the stabilizing background may include acquisition, processing, diachrony, discourse pressure, and literate or institutional transmission. Measurement design supplies the audit trail, not a stabilizer. None of those should be collapsed into a universal language-internal category.

Truth-tracking supplies a different portability test. The concrete prediction is contrastive: self-consistency or fluent coherence should help where coherence is the missing support, while retrieval or tools should help only when they strengthen the needed stabilizer. That contrast follows if representational success depends on partially coupled routes such as perception, testimony, coherence, measurement, intervention, error-correction, and practical success (Reynolds, 2026b). A large language model can participate strongly in some routes and weakly in others; surface improvement and truth-tracking improvement can come apart.

A simple evaluation profile would pre-register the missing support. If answers to document-grounded questions fail by making unsupported claims, retrieval should improve citation-grounded accuracy where self-consistency only improves fluency. If retrieval helps only when the supplied passages are relevant and used, it's earned its role. If fluent agreement rises while unsupported claims remain, the profile is coherence-stabilized rather than truth-tracking in that setting.

In each case, the same vocabulary applies without imposing the same mechanism. The deital profile is grammatical and distributional. The typological profile is comparative and methodological. The truth-tracking profile is epistemic and representational. Calling all three PROFILES doesn't erase those differences. It lets us ask the same disciplined questions: what gets projected, what pattern supports it, what stabilizes it, and where it fails.

Fields, practices, and institutions should be handled in the same way. They can be scope conditions or stabilizing backgrounds; in some social cases, institutional facts may also belong to the profile itself. Fields and practices should not be promoted to kinds merely because they organize inquiry or action. A field may condition which projections matter without itself being the projection target. A practice may stabilize a category without being the kind. Those are bridge claims, and they need their own evidence.

5 Failure Modes and Demotion

The vocabulary is useful only if it can say no. Without failure modes, projectibility profile would become another loose label for any interesting cluster. A profile view needs demotion rules: ways to say that a category is too thin, too local, wrongly levelled, or overpromoted.

The first failure mode is a thin profile. A thin profile has diagnostics that identify familiar cases but don't license further expectations. The category may still be convenient, but the kind claim is weak. In grammatical work, a label that sorts textbook examples but predicts nothing about mismatch, manipulation, acquisition, or variation is thin. In typology, a comparative label that cannot generalize beyond the coding sample is thin.

The second failure mode is wrong-level projection. This happens when evidence at one level is used to project at another without a bridge. A corpus distribution doesn't by itself establish a model-internal mechanism. A social-practical regularity doesn't by itself establish a biological disposition. A grammatical category in one language doesn't by itself establish a cross-linguistic comparandum. The projection may be possible, but the bridge has to be argued.

The third failure mode is decorative mechanism. A paper names a mechanism, but the mechanism doesn't underwrite any new projection. The test from Section 3 returns here as a demotion rule: if removing the mechanism talk leaves the same predictions, the mechanism has not earned its place in the kind claim.

The fourth failure mode is over-homeostatization. Stability, network order, and maintenance are often real achievements. They should not be relabelled as homeostasis unless the evidence shows corrective organization preserving a higher-scale relation through lower-scale variation. The risk is especially high outside biology, where the attraction of the homeostasis metaphor can hide weak evidence. A maintained convention, a stable distribution, or a reliable measurement practice may be projectible without being homeostatic.

The fifth failure mode is scope overreach. A profile can be real and useful under one set of conditions while failing elsewhere. The error is to export the projection without fresh tests. A category projectible for discourse analysis may not be projectible for syntax. A model behaviour projectible under benchmark conditions may not transfer to tool-using settings. A social status projectible inside one institution may not travel across institutions.

Demotion belongs inside realism. If kind claims are answerable to the projections they support, then failed projection should change the claim. Sometimes the right response is rejection: the category doesn't support the advertised inference. Sometimes localization is right: the category is projectible only under narrower scope conditions. Sometimes relevelling is right: the pattern is real, but at a different explanatory level.

Profile talk joins promotion and demotion. A category earns stronger status when its profile supports reliable projection, when any claimed stabilizers are identified at the right level, and when its limits are known. It loses status when those conditions fail. The framework therefore gives anti-essentialist kind theory an exit procedure, not just a route into realism.

6 Conclusion

The slogan is compact: a category is a profile, stabilized by relations or mechanisms, projectible under specified conditions. Each clause matters. A category is a profile because no single essence need define it. Stabilization matters because recurrence needs explanation. Projectibility matters because kind claims earn their scientific or grammatical value by licensing further expectation.

Homeostasis remains important, but it no longer has to carry every achievement. Two axes have to stay apart. Network order is a structural claim about why co-availability isn't coincidence. Stabilization, maintenance, and control are modal or diachronic claims about persistence, recurrence, and return. A profile can be strong on one axis and weak on the other. Only the controller case deserves the strict homeostatic label. Keeping those distinctions apart preserves Boyd's anti-essentialist realism while incorporating Slater's stability-first lesson and Khalidi's network discipline.

The result is a portable vocabulary for asking better questions. What does this category let us infer? What pattern supports that inference? What stabilizes the pattern? What would defeat the projection? And when should the category be demoted, localized, or rejected? Those questions are the work a theory of kinds has to do.

The framework doesn't settle every grounding question. It doesn't decide in advance whether a projection is grounded causally, conventionally, normatively, or by some mixture of supports. It doesn't solve Goodman's new riddle. And it doesn't assume that all six profile questions will receive strong answers in every case. It gives the order in which those answers have to be earned.

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